Resource Extraction and Conversion from Lunar Anthropogenic Inputs with Microwaves

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1 Vision and Innovation

1.1 Brief overview of solution, eligible trash items to be recycled, overview of recycling system, and outputs (both usable and unusable)

The Resource Extraction and Conversion from Lunar Anthropogenic Inputs with Microwaves (RECLAIM) system is a modular lunar recycling system that *processes* and *repurposes* metallic and plastic waste streams into useful artifacts and other feedstocks using high-power microwaves. RECLAIM plugs into the Infrastructure for Lunar Utilization of Microwave Energy (ILUMEN) system proposed by Bilén [1], which is an extensible infrastructure to leverage multiple-use, high-power microwave sources for a range of integrated materials processing, services, and other critical needs on the Moon.

As shown in Figure 1, the RECLAIM system's recycling process begins with the physical conversion of anthropogenic waste streams (e.g., metals, plastics, foams) via a shredder into smaller pieces that can be processed within RECLAIM's microwave reactors. In the current iteration of the system, the shredded pieces are placed manually into the reactors, but this can be automated in a future lunar version. RECLAIM is connected to ILUMEN's high-power microwave source, and a waveguide selector directs power to one of two reaction chambers, one for smelting metals and the other for pyrolyzing plastics.

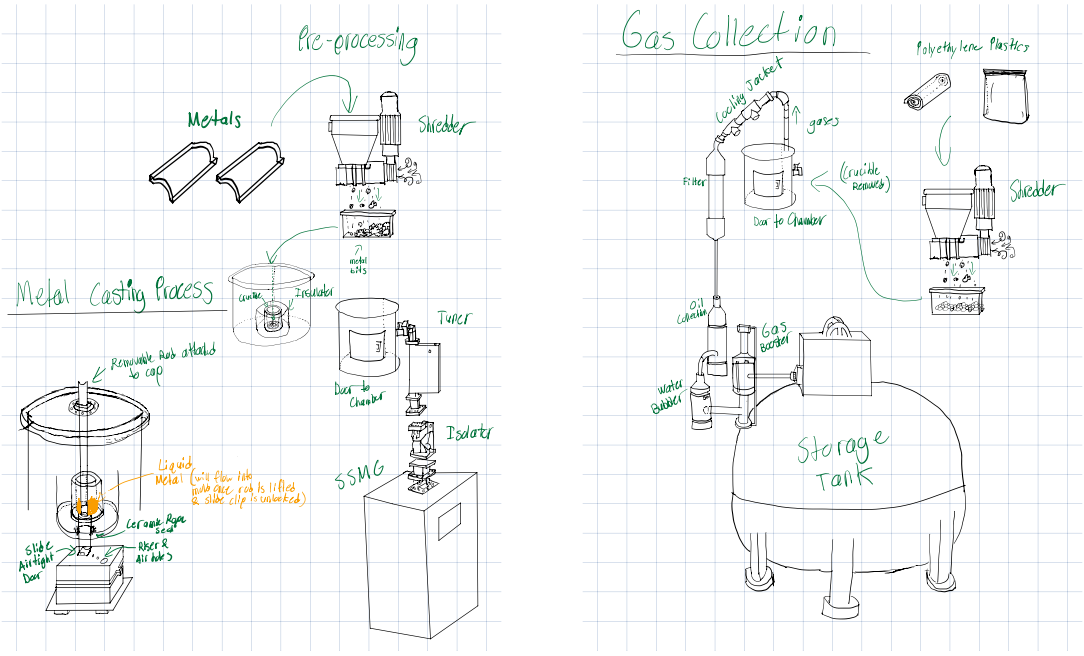


Figure 1: High-level overview of RECLAIM processes.

The metal (i.e., smelting) reactor smelts metallic waste so that the casting process (via investment and dedicated molds) converts it into usable artifacts such as tools or replacement parts. The plastics (i.e., pyrolysis) reactor converts plastic and other polymer feedstocks (e.g., various plastics, foams) into gaseous precursors for fuel and plastic/carbon char for uses such as filtration systems and fillers for lunar regolith-based concrete. Other outputs from plastic reformation include waxes and oils that are useful for other purposes.

The eligible trash items that we will recycle in this challenge include aluminum structure/struts, reclosable bags, and drink pouches. All items will be preprocessed using the shredding process to

convert them into small pieces and will not be chemically changed. For metal, once preprocessed, the metal chips will be smelted in a reactor and the molten metal made into the desired artifacts via casting. The shredded plastics will be converted by the microwave pyrolysis reactor, which breaks down the plastics into syngas, pyrolysis oils, plastic char, and wax. Additionally, since the drink pouches are composed of 24% aluminum, this will reduce to aluminum nodules that subsequently can be smelted in the metal reactor and cast. During the LunaRecycle Challenge, we may also attempt using RECLAIM to reprocess other trash items within its plastics reactor: gloves, Nomex bags, Zotek F30 foam, rehydratable pouches, and clothing are all processable through the pyrolysis reaction with various adjustments to the reactor. RECLAIM inputs are not constrained to this list, as it can smelt and melt lunar in-situ resources such as ilmenite and process a variety of other pyrolyzable products (e.g., fabrics).

1.2 How RECLAIM uses innovation to address the Challenge

The RECLAIM system enables lunar recycling and processing of materials into new artifacts through its use of ILUMEN's centralized high-powered microwave source. ILUMEN enables multiple applications to utilize microwave power by tapping into a centralized infrastructure rather than each application requiring its own source. This will enable the scale up of lunar exploration and commerce through an extensible infrastructure that leverages multiple-use, high-power microwave sources for an array of integrated lunar materials processing, recycling, services, and other needs. This method will decrease the total mass requirement for transported equipment from Earth to the Moon.

The top-level system architecture depicted in Figure 2 shows the primary components of ILUMEN, including how it interfaces with external systems. RECLAIM is, hence, just one of many potential ILUMEN applications addressing many of the needs supporting a sustained lunar presence.

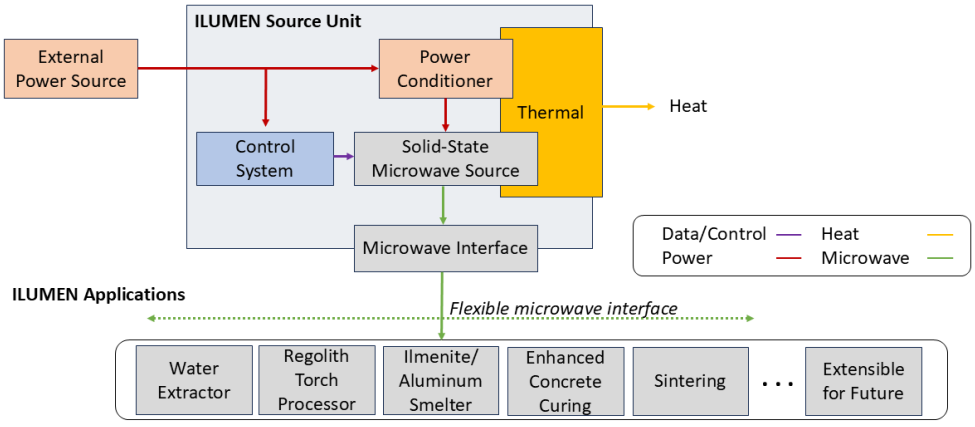


Figure 2: Systems diagram showing ILUMEN's multiple-use microwave source processing of lunar feedstock materials for construction applications, extensible to future applications.

1.3 How RECLAIM builds and improves upon current state of the art

RECLAIM builds upon existing microwave heating and terrestrial smelting technologies by extending them into a highly efficient, modular, and semi-autonomous system optimized for repurposing metallic and plastic waste streams. Whereas microwave heating of lunar materials such as ilmenite has been demonstrated in laboratory and experimental settings, RECLAIM advances the state of the art by enabling reliable, repeatable heating of both metallic materials, such as aluminum, and plastic waste, such as polyethylene, within a single system. The use of a silicon carbide crucible to convert microwave energy into thermal energy overcomes traditional limitations associated with direct microwave–metal interactions, allowing higher temperatures, improved energy efficiency, and broader material compatibility. The modularity of the system allows for a separate process to pyrolyze plastics and use the byproducts for fuels, filters, and fillers in concrete.

RECLAIM's microwave reactors employ cylindrical geometry because they are designed to produce a uniform electromagnetic field distribution within the cavity, unlike rectangular geometries in which the field varies volumetrically [2]. Microwave heating is more efficient than conventional heating methods because the system requires less energy to transfer more heat, thereby decreasing reaction time and power consumption. Microwave heating couples directly with materials, achieving more uniform internal heating in a controlled atmosphere through electromagnetic fields. Microwave technology is widely recognized for its use in cooking appliances. Significant advances have extended its application into metal processing [3], plastic pyrolization, and manufacturing. For RECLAIM's metal processing subsystem, molten metal is poured into a prepared castings and molds. For RECLAIM's pyrolysis subsystem, syngas, oil, and plastic char are collected for a variety of uses on the lunar surface. Collectively, these innovations position RECLAIM as an innovative lunar recycling and manufacturing platform.

Within RECLAIM, the metal subsystem is a simple and innovative way to recycle a variety of different metallic compounds including aluminum and in-situ resources at a lower energy cost. RECLAIM can recycle $> 95\%$ of metal scrap (theoretically 100%) with high energy efficiency and requiring only electrical power. It achieves temperatures of 1300°C or higher. Future iterations of the RECLAIM system could be more fully automated to reduce crew time during lunar missions.

The RECLAIM pyrolysis subsystem was directly integrated into the original SMELT system that was developed during the 2023 NASA BIG Idea Challenge. This system expands the capabilities of the SMELT system by introducing subsystems that allow for the processing of plastic waste. The new pyrolysis subsystem maintains similar power levels and efficiency, while also minimizing mass by combining two systems into one modular design.

1.4 How RECLAIM leverages advanced technologies

RECLAIM uses a solid-state microwave generator (SSMG) provided by ILUMEN to enable efficient in-situ material processing and recycling on the lunar surface. The SSMG provides microwave power at 2.45 GHz (this is a common ISM frequency used, for example, in microwave ovens). Solid-state microwave technology has made substantial progress recently. The new technology of gallium nitride (GaN) transistors allows for higher efficiency and power densities over older gallium arsenide (GaAs) transistors, enabling solid-state sources to become more compact and offer higher power outputs than previously. SSMGs also have several benefits over conventional magnetron microwave generators. Solid-state sources are faster and more precise, use low-voltage DC power instead of high-voltage power, and have much longer lifetimes. The modular design of many solid-state sources also enables easier troubleshooting and repairs than magnetrons, which have sensitive parts like magnetron tubes that are fragile and difficult and costly to service.

For comparison, terrestrial recycling approaches such as melting in electric arc furnaces require high power levels and oxidizers to function properly. These systems consume large amounts of energy and have significant infrastructure requirements. In contrast, RECLAIM recycles metals using only electrical power and a cooling system while achieving over 95% material recovery at significantly lower energy cost.

RECLAIM is inherently suited to vacuum and reduced gravity environments since microwave heating does not rely on combustion or convection systems, which would require oxidizers and complex cooling systems on the lunar surface. The SSMG directly couples electromagnetic energy into the silicon carbide crucible and maintains stable heating under extreme temperature swings, abrasive lunar dust, and limited power budgets. Similarly, plastic material is heated both without a crucible and only requires in-situ catalysts. By providing a lightweight, power-efficient, and semi-autonomous microwave forge, RECLAIM enables sustainable material reuse, reduces dependence on Earth resupply, and supports the production of hardware, tools, spare parts, utensils, structural components, and gases directly from recycled material on the lunar surface.

1.5 Commercial potential of RECLAIM

RECLAIM has strong commercial potential because of its ability to recycle multiple waste streams. ILUMEN creates the lunar "microwave economy" [1], which RECLAIM becomes part of as an additional user (buyer) of its service and provides back artifacts (seller) for users on the Moon.

The modular system greatly reduces the mass and power required to recycle both metal- and plastic-based materials by combining them into one system. The microwave-powered recycling system has strong commercial potential due to its ability to convert plastic waste into valuable gas and char byproducts and reform metals into desirable parts. Unlike traditional recycling methods that rely on large facilities, this solution enables processing of waste in more far-reaching areas, reducing transportation costs and allowing waste to be managed closer to its source. This makes the system useful for municipalities, industrial sites, and organizations seeking more flexible and efficient recycling options.

The system is particularly well-suited for industrial and remote applications where both metal and plastic waste disposal and energy supply are challenges. Manufacturing facilities, military installations, and remote communities could use the system to reduce waste while re-purposing broken parts and extracting fuel products for supplemental energy or other purposes.

Additionally, due to the growing regulatory and economic pressure around the world to reduce plastic waste, our technology's commercial viability extends even further. The pyrolysis subsystem is a closed-loop recycling system, meaning 100% of the plastics within the system are processed and recycled, thereby reducing landfill sizes. Additionally, the system could help companies meet emission and recycling standards regardless of how far a recycling plant is. The system's capability for re-purposing both in-situ resources and waste metals allows for low-cost product production through casting. The modular design allows the system to be scaled or integrated into existing waste-management infrastructure, making it adaptable to a wide range of customers and use cases. With additional refinement, this solution could serve as an efficient, commercially viable approach to decentralized recycling and byproduct extraction.

1.6 Anticipated maintenance needs and expendables required for long-term operation of RECLAIM (one year or longer)

For long-term operation (over one year), the system's maintenance needs are expected to include components exposed to high temperatures, as well as components that regulate the pyrolysis environment. Routine inspection and replacement of seals and other fittings will be required to maintain their operations and prevent leakage of gases. Sensors and electrical connections will be verified as operational at startup to ensure consistent performance and system safety.

The catalysts used in the gas cracking stage are viewed as expendables as they will degrade over time. These catalysts are designed to be accessible and replaceable with in-situ resources such as ilmenite from lunar regolith, which minimizes the need for replenishment from Earth. Components used to separate oil and gas byproducts may also require periodic cleaning or replacement to prevent clogging and maintain efficiency.

To optimize maintenance times, the system is designed with modular subsystems that allow individual sections and components to be serviced or replaced without requiring the entire system to be disassembled. Preventive maintenance procedures established during testing would also be followed to ensure reliable long-term operation. Overall, the anticipated maintenance and expendables are limited and manageable even for long-term operation.

2 Milestone Round Solution Design

2.1 Milestone Round Engineering Design

2.1.1 Rationale for design approach

The proposed RECLAIM recycling system employs a shredding mechanism to preprocess both metallic and plastic materials that are placed into reactors to recycle products into reusable outputs with minimal waste. The processes include a plastic pyrolysis reactor to break down plastics into gaseous products, such as syngas, that can be upgraded and utilized later as chemical precursors, and a reactor to melt aluminum for casting artifacts.

The foundation of the recycling system is a reliable and efficient electrically-based heat source. Microwave heating was chosen due to its many advantages over conduction- or convection-based heating systems. Microwave heating employs volumetric heating, meaning it penetrates into the material and heats it more evenly. By heating the materials directly and ensuring the heat is evenly distributed throughout, the heat-up time is lower, and the overall power consumption decreases. Consequently, microwave heating results in a more efficient form of heating to both smelt aluminum and pyrolyzing plastics. Another benefit of microwave heating is the selective heating behavior. Certain materials, such as plastics or silicon carbide, absorb microwaves, whereas others, such as metals tend to reflect them. By reflecting radiation, the reaction chamber will experience lower thermal loading while the plastic material and the silicon carbide crucible containing the aluminum can still be heated to the necessary temperatures for pyrolysis and smelting, respectively.

Plastic pyrolysis is an effective option to recycle undesirable and excess plastic waste into valuable, usable resources. Transporting materials and consumables to the Moon is very expensive. Long-term lunar missions will have plastic waste build-up. RECLAIM offers the ability to reduce pollution and lunar "landfills" by re-purposing these valuable resources. The pyrolysis process converts plastics into valuable materials such as plastic char and syngas. These byproducts are precursors for several applications on the lunar surface, including the production of rocket propellant, filtration systems for lunar habitats, fillers for concrete, and other gas-based fuel sources [4].

Polyethylene plastic is comprised of repeating units of ethylene (C_2H_4) that form extremely long hydrocarbon chains. These long hydrocarbon chains act as the starting material for the pyrolysis process. The reaction begins when heat is added to these hydrocarbon chains. The heat causes chain scission reactions to occur, leading to the degradation of these long chains. C–H bonds tend to break first, followed by the C–C bonds as the temperature increases. The chains continue to degrade further as the reaction progresses until volatile gases are formed [5]. Volatile gas products that are produced during this process include hydrogen gas (H_2), methane (CH_4), ethylene (C_2H_4), propene (C_3H_6), etc. These volatile gases form via reactions with radicals and small hydrocarbon compounds after the chains have degraded. The highly reactive unpaired electrons in the radicals make these reactions possible. These chemical reactions are provided below.

$$\text{Polyethylene Carbon Chain} \xrightarrow[\text{(Heat)}]{\Delta} \begin{array}{l} \mathrm{C_2\,H_4 + CH_3^{\cdot}} \\ \mathrm{C_3\,H_8 + CH_3^{\cdot}} \\ \mathrm{C_2\,H_5 + H^{\cdot}} \\ \mathrm{C_3\,H_7^{\cdot}} \end{array}$$

$$\begin{aligned} \mathrm{C_2\,H_4 + CH_3^{\cdot}} &\longrightarrow \mathrm{C\,H_4 + C_2H_3^{\cdot}} \\ \mathrm{C_3\,H_8 + CH_3^{\cdot}} &\longrightarrow \mathrm{C\,H_4 + C_3H_7^{\cdot}} \\ \mathrm{C_2\,H_5^{\cdot} + H^{\cdot}} &\longrightarrow \mathrm{C_2\,H_4 + H_2} \\ \mathrm{C_3\,H_7^{\cdot}} &\longrightarrow \mathrm{C_3\,H_6 + H^{\cdot}} \end{aligned}$$

Pyrolysis takes advantage of the materials from which plastics are made, returning them to their initial state. This not only manages waste but also contributes to ISRU-driven mission sustainability.

For the Milestone system design, the pyrolysis reaction takes place within a proof-of-concept system (Figure 3) that is independent of the metal smelting system. In order for the reaction to take place, the entire pyrolysis system is purged with argon gas to produce an inert atmosphere. This reduces the oxygen concentration below 1% to have a suitable environment. (Note, any inert gas will work, and a potential lunar system could reclaim and reuse these gases.) Following this purging process, both the microwave system and the cooling system are turned on, at which time the pyrolysis reaction begins in the reaction chamber.



Figure 3: Proof-of-concept microwave plastic pyrolysis system.

The next phase of the plastic pyrolysis process is to cool down the resulting gases using a cooling jacket. The milestone design uses a black steel pipe that concentrically surrounds the pyrolysis-

gas-effluent pipe. Cooling water runs through the surrounding pipe to rapidly cool the pyrolysis vapors, condensing them into oil and leaving noncondensable gases to continue flowing. Water was selected due to its high heat transfer capability and operational simplicity, and is used in a closed-loop cooling system, so it is not an expendable.[1] The cooling jacket reduces the temperature of the gases to around 27–38°C, ensuring that the condensable gases are converted to oil and the noncondensable gases are cooled to a storable state. A T-connector is used to separate the oil and gas streams, forming a simple, gravity-based separation subsystem.

The milestone metal recycling system is designed to efficiently melt metals to be transferred into an investment casting or mold to be formed and re-purposed into a usable artifact. Figure 4 depicts the proof-of-concept setup of the metals recycling system. Unlike plastic materials, metal does not readily absorb microwave energy, reflecting the microwaves instead, making direct heating of the metal difficult. As a result, the smelting process requires the use of a microwave-absorbing crucible. The crucible absorbs the microwaves and transfers the heat directly into the metal through conduction. To meet this requirement, a silicon-carbide (SiC) crucible was selected because it readily absorbs microwaves and stores heat very well. The melting point of SiC is $\sim$ 2700°C, which is well above the melting point of aluminum and most metals. SiC also has high thermal conductivity, which reduces thermal gradients, minimizing hot spots and promoting uniform heating of metals. SiC also offers high resistance to thermal shock, tolerating cycles of rapid heating and cooling without cracking. Furthermore, the SiC crucible is surrounded by ceramic fiber wrapped in dielectric tape, which acts as an insulator for the crucible. Ceramic fiber and dielectric tape were chosen for many reasons, the first being their compatibility with microwaves. The insulation has a low dielectric loss factor, meaning it does not absorb microwaves and lets them pass directly to the crucible. Additionally, ceramic fiber has low thermal conductivity, which means it does not absorb heat from the crucible. Finally, ceramic fiber does not melt, decompose, or chemically degrade at temperatures $>$ 1200°C.



Figure 4: Proof-of-concept of metal recycling system.

2.1.2 Preliminary evidence and analysis predicting performance including mass, volume, and power requirements

Research on microwave-driven pyrolysis of plastics has shown that polyolefin feedstocks such as LDPE, HDPE, and PP can produce a gas-dominant yield under appropriate power, temperature,

[1] Indeed, as part of the ILUMEN paradigm, this heat could be used in external systems, such as heating habitats and other critical systems.

and absorber conditions [6]. Non-catalytic microwave pyrolysis of LDPE yields approximately 70–75% gaseous products while only producing 20–25% oils and less than 5% solids. This indicates a bias toward permanent gas formation during pyrolysis under the proper conditions. However, at reaction conditions initially tried at 500°C, the wt% of product was closer to 79% oils and 21% gases. Heating to much higher temperatures such as 2900 K, can favor gas production, specifically for hydrogen and methane. At higher temps, hydrogen can be expected to make up 59% of all gases produced. Expected gas products include hydrogen, ethylene, methane, ethane, and propene.

Although these milestone experiments were conducted with a chamber temperature around 450–500°C, the current pyrolysis system heats plastics to approximately 300–600°C, which sits beneath the gas-yield optimum reported for many other pyrolysis systems. At this temperature with microwave-assisted operation, cracking reactions within the plastics can still proceed. Prior studies show that increasing power can compensate for a lower temperature by accelerating the bond-breaking process, favoring the formation of non-condensable gases over liquid products. As microwave power increases, gas yield also increases while oil yield decreases. Additionally, including a catalyst within the chamber will increase gas yields. Based on literature, the proposed system operating at 300–600°C with the addition of a catalyst within an external heating jacket heated to 800°C will preferentially generate non-condensable gases.

The metal smelting system is designed to melt ~ 0.5 kg of aluminum per operating cycle using microwave heating at a frequency of 2.45 GHz. Aluminum is heated to higher than its melting point of 660°C. As melting aluminum does not involve a chemical transformation, the recovered aluminum mass closely matches the initial mass, minus minor losses due to handling and surface oxidation.

Prior to testing, a lower-bound energy requirement for melting aluminum can be estimated using first-principles thermodynamics. The total heat transferred, Q_{total}, can be found by first finding the heat transfer required to taking aluminum from solid-to-liquid state, Q_{solid}, calculated by

$$Q_{\text{solid}} = m_{\text{material}} c_{\text{solid}} (T_{\text{melt}} - T_{\text{initial}}), \tag{1}$$

where m_{material} is the mass of the aluminum, c_{solid} is the specific heat capacity of solid aluminum, or the amount of energy required to raise the temperature of one unit mass of aluminum by one °C at a constant pressure, T_{melt} is the melting point of aluminum, and T_{initial} is the initial temperature.

Once the aluminum reaches the melting point, the latent heat of fusion must be factored in. The heat transfer as the aluminum changes phases, Q_{fusion}, can be calculated using

$$Q_{\text{fusion}} = m_{\text{material}} L_{\text{f}}, \tag{2}$$

where L_{f} is the latent heat of fusion of aluminum, or the amount of energy the material must be absorbed to change phases. For simplicity, the assumption will remain that aluminum does not absorb any more heat after reaching the melting point. If this were not the case, the specific heat capacity of liquid aluminum would need to be factored in as the material would continue to absorb energy from the crucible. After calculating both Q_{fusion} and Q_{solid}, the two can be added to obtain the total heat transfer into the aluminum during the smelting process.

Assuming a specific heat capacity for solid aluminum of 900 J/kg·K, latent heat of fusion of 397 kJ/kg, initial temperature of 25°C (room temperature), and final temperature of 660°C, the process of smelting 500 g of aluminum requires ~ 484 kJ of applied energy. This value can then be converted into kilowatt hours (kWh) when divided by 3,600, yielding 0.136 kWh. The total power, P_{tot}, can then be found via

$$P_{\text{tot}} = \frac{Q_{\text{tot}}}{t} \tag{3}$$

where t is the reaction time in hours. While this reflects the theoretical energy required, realistic system have conversions efficiencies that must be accounted for.

A similar analysis can be performed for the plastic pyrolysis system processing polyethylene, a common substance found in lunar packaging waste. Before calculating the total energy required, an important distinction must be made. When calculating for aluminum, it was assumed that the heating stopped at the melting point. However, for the plastic pyrolysis, the chamber will reach roughly 300°C. To account for this, another equation must be introduced to calculate the total energy transfer while in liquid phase, i.e.,

$$Q_{\text{liquid}} = (m_{\text{material}})(c_{\text{liquid}})(T_{\text{final}} - T_{\text{melt}}), \tag{4}$$

where c_{liquid} is the specific heat capacity for polyethylene and T_{final} is the final temperature that the material within the chamber reaches. After calculating Q_{solid}, Q_{fusion}, and Q_{liquid}, and finding the sum, a lower bound energy requirement estimate of 806.3 kJ is found, or 0.224 kWh.

2.1.3 System-level and component-level design specifications for hardware and software, including bill of materials

In the Milestone system, the reactors are energized by a Daihen ATM15a 1.5-kW magnetron connected to a Daihen CMC-10A Automatic Microwave Tuning Control Unit. The magnetron was controlled by a Daihen ATP-10A. The system uses a WR340 isolator connected to a Daihen SMA15B tuner to match the network to the resonance chambers. The isolator subsequently reduces reflected microwaves back into the source. The milestone bill of materials is given in Table 4.

Table 1: Milestone System Bill of Materials.

Item #	Part Name	Manufacturer	Part Number	Qty
1	Mass Flow Controller	MKS Instruments	1579A	1
2	Multi Gas Controller	MKS Instruments	647B	1
3	3/4 in x 18 in Black Steel Pipe	PLUMBFLEX	3418PBL	1
4	3/4 in x 24 in Black Steel Pipe	PLUMBFLEX	3424PBL	1
5	1 1/4 in x 12 in Black Steel Pipe	PLUMBFLEX	1118PBL	1
6	3/4" 45 deg. Elbow FNPT MNPT	SOUTHLAND	520-504HN	2
7	1.5 kW magnetron	Daihen	ATM15a	2
8	Microwave Tuning Control Unit	Daihen	CMC-10A	2
9	Magnetron Controller	Daihen	ATP-10A	2
10	Isolator	Daihen	WR340	2
11	Waveguide Connector	N/A	WR340-WR340	1
12	Waveguide Connector	N/A	WR430-WR340	1
13	Waveguide Connector	N/A	90 deg WR430-WR430	2
14	1-1/4" x 3/4" Tee Fitting	SOUTHLAND	520-664HP	2
15	3/4" 90 deg. Elbow FNPT	STZ	PDB E90-34	2
16	1 1/4" Black Steel Pipe Nipple	STZ	300 114XCL	2
17	1/2" x 10" Black Steel Pipe	PLUMBFLEX	GS1/210BTBE	2
18	1/2" x 6" Black Steel Pipe	PLUMBFLEX	GS1206BTBE	2
19	1/2" 90 deg. Elbow	SOUTHLAND	520-003HN	2
20	1 1/4" x 1" Reducing Coupling	SOUTHLAND	511-365HN	2
21	Water Hose 3/4"	Teknor	S-26122	2
22	Gas Hose 1/2"	Westinghouse	W12NGH25	1
23	Argon Cylinder	N/A	N/A	1

The microwave plastic pyrolysis system utilizes the same source and control units to manage the energy input. The pyrolysis system also utilizes the Multi Gas Controller 647B from MKS Instruments and a Mass Flow Controller to manage the input of argon into the system. The team fabricated a cooling jacket using concentric NPT black steel pipes and malleable iron fittings. Hoses were attached to the inlet and outlet of the cooling jacket, which received and returned water from the building. Level zero system-level requirements are depicted in Figure 5. As we focused mostly of Milestone proof-of-concept systems, a significant early part of the Final Round preparation will occur in a full and complete set of system requirements.

Competition Level Requirements	System Level Requirements	Subsystem Level Requirements	Component Level Requirements
Recycle Aluminum Waste	Heating via Microwave Energy	Cylindrical Reaction Chamber	Thermally Resistant
Recycle Plastic Waste	Argon Purge <1% O2	1.5kW >= Power System	Minimal Maintenance Required
	Cooling Jacket Functionality	Multi Gas Controller	Airtight
	Oil and Gas Collection	Side Door for Access	Swagelok Piping
	Solid byproduct collection/casting.	Syngas Collection	
		Mass Flow Controller capable of >3SLM	
		Access to cool water	

Figure 5: Level zero system requirements.

2.2 Milestone Round Operations, Analyses, and Testing

2.2.1 Concept of operations for one cycle and any related analyses or testing, including thermal analysis and structural analysis, if relevant to your solution

During the Milestone Round operation, RECLAIM is comprised of two functionally independent systems: the smelt system and the pyrolysis system. Both systems utilize a shared microwave-based power source to provide the required thermal energy for processing. With respect to applications of microwave energy, the operational flow diverges into the respective smelting and pyrolysis processing paths, which proceed independently to produce their designated outputs. This concept of operations describes the nominal operational sequence and material flow associated with these processes under normal operating conditions. The full concept of operations is illustrated in Figure 6.

In the smelt system, a crucible is filled with pre-processed aluminum and placed inside the microwave chamber. Microwave energy is used to heat the crucible until the aluminum becomes fully molten and ready for casting. During this process, the chamber remains sealed, and no gases are produced that require handling. Once the aluminum has melted, microwave power is shut off, the chamber is opened, and the crucible is removed and poured into a sand mold for casting.

In the plastic processing system, the same microwave chamber is used, but it operates under different conditions. The chamber is loaded with plastic feedstock along with a carbon-based material to improve microwave absorption. An inert gas such as argon purges the system to create suitable reaction conditions. Microwave power is then applied to drive the pyrolysis reaction. As the plastics decompose, gases are produced and flow out of the chamber through dedicated piping. These gases pass through a cooling section where condensable vapors are collected as liquid oils, while non-condensable gases are vented or handled separately. Solid residues remain in the chamber and are removed after the reaction is complete.

Although both systems use the same microwave technology, they currently operate as separate processes and do not run at the same time. The smelt system is focused on melting and casting metal, whereas the plastic system is focused on chemical decomposition and gas handling. By using the same power source for two different processing modes, the overall system remains simple while still allowing both aluminum recovery and plastic conversion to be performed efficiently.

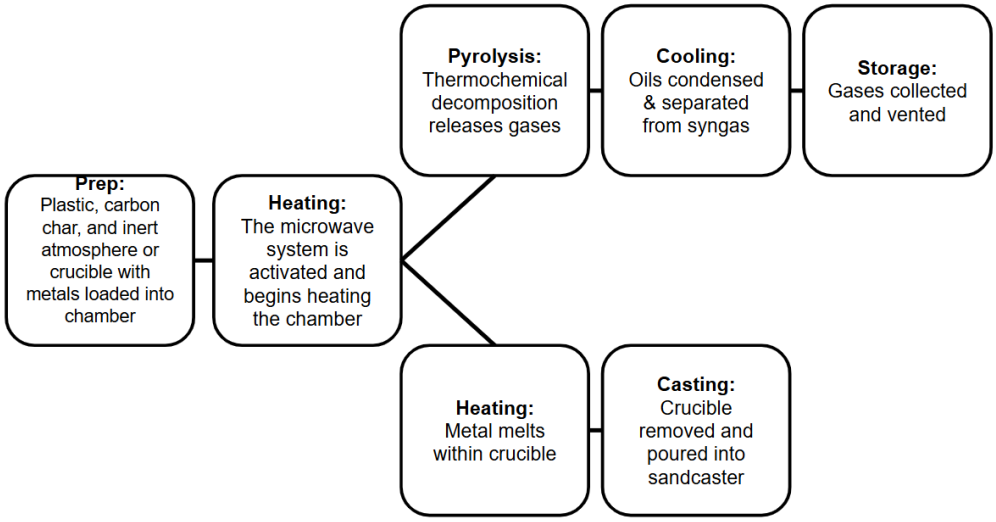


Figure 6: Concept of Operations

An analysis of the resonance chamber was conducted using ANSYS HFSS, a 3D electromagnetic simulation software. Preliminary research was conducted to determine potential geometries for the resonance chamber. A cylindrical design was shown to eliminate hot spots found in rectangular chambers. A cylindrical design also distributes stress from heat more evenly, meaning the chamber will experience less stress concentrations, keeping the observed stress below the material's failure threshold [7].

ANSYS HFSS modeling reveal further advantages of the cylindrical geometry, showing high electric fields at the center of the chamber. After further ANSYS simulations, a model with a 16-cm radius and a 26.4-cm height was selected. These dimensions ensure a minimization of the reflection coefficient, to minimize power loss due to reflections, and the E-field in the heating zone was maximized, optimizing heat transfer to the crucible. The ANSYS HFSS simulation results are shown in Figure 7.

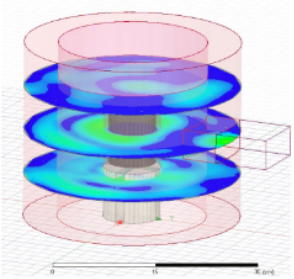


Figure 7: ANSYS HFSS simulation of resonance chamber.

For the SMELT precursor system, thermal experiments were conducted to determine the thermal energy sufficient to increase the material temperature to the desired temperature at which a complete phase change occurred. Figure 8 depicts the graph of recovery of the metal vs. the total transmitted energy.

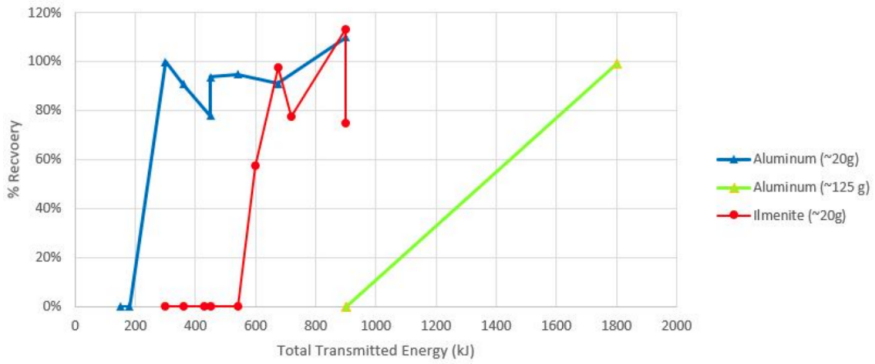


Figure 8: Total recovery vs. total energy for various metals.

Tests were run on small batches of aluminum and ilmenite and a large batch of aluminum. The data show that the percent metal recovered has a direct correlation with the total energy delivered to the system. Additionally, it is reasonable to assume that as more energy is added to the system, melt time will reduce. Overall, the smelt system allows faster melt times and a higher metal recovery rate, scaling with energy input.

2.2.2 Summary of the inputs, outputs, and efficiency of the recycling process

Table 2: RECLAIM Recycling Summary.

Question	Output
Eligible trash item(s) from TABLE 4 that RECLAIM will recycle	Aluminum structure/struts Drink Pouch Rehydratable Pouch
Total mass of one batch of eligible trash items (expressed in kg).	1.20 kg Meets MBR by > 500 g and via makeup
Mass of each eligible trash item in one batch (expressed in kg).	0.50 kg of Aluminum 0.35 kg of Drink Pouch 0.35 kg of Rehydratable Pouch
Length of time to complete one cycle	1.25 hours
Electricity required for one cycle	Peak Microwave Power = 1.5 kW 40% efficiency assumed Total Electricity = 1.5 kW / 0.4 = 3.75 kW Total Power = 3.75kW * 1.25 hrs = 4.69 kWh Net Electricity = 0
Amount of water required for one cycle.	4 kg Closed Loop Cooling System
Types and amounts of other inputs required for one cycle. List each additional input and the mass required for one cycle	50 g activated carbon for heating plastics
Number of crew and time per crew member required to operate one cycle	1 crew member 0.25 hours of operation per cycle.
Usable outputs produced by one cycle and amounts of each	Reusable 3DCP nozzle: 0.490 kg. Pyrolysis gas as fuel or propellant: 0.15 kg Plastics char used as a filler for concrete: 0.025 kg Pyrolysis oil used as a fuel source: 0.50 kg
Unusable outputs produced by one cycle and amounts of each	0.035 kg Lost Metal via casting and undesirable waxes from pyrolysis reaction
Calculate Mass Efficiency for one cycle	97%
Calculate Energy Efficiency for one cycle	0.23 kg/kWh

2.2.3 Plan for characterizing usable outputs and finished products from the recycling process and verifying this characterization

The smelt system produces a primary usable output in the form of molten metal using aluminum feedstock. Characterizing this output is essential to verify the system's effectiveness for recycling, manufacturing, and its ability for ISRU. This characterization plan focuses on quantifying material recovery efficiency, assessing the output quality, and verifying the repeatability across multiple cycles.

The main metric for characterizing the smelt system output is mass recovery. For each cycle, the mass of the input material will be recorded prior to smelting, and the mass of the cast metal after cooling will be measured. This enables direct calculation of recovery efficiency and material losses. Test results demonstrate recovery efficiencies $> 95\%$ by mass, and additional trials using similar aluminum inputs will be conducted to further verify that the high recovery is consistent and reproducible. Apart from the mass recovery, the physical quality of the recovered metal will be evaluated. Observations such as surface finish, uniformity of the melt, and presence of residual contaminants will be documented after each run. Figure 9 shows one of the successful smelting tests.

The finished products that the melt system can create, such as ingots, custom parts, or simple tools has been proven through testing. The system has successfully cast both ingots and a nozzle for a 3D printing concrete company. The ability to successfully cast and solidify the inputs into usable forms demonstrates that the material retains sufficient integrity for manufacturing applications. Repeating this process across multiple cycles confirms that the SMELT outputs can withstand repeated reuse and will not be degraded by the process.



Figure 9: Melted aluminum in the crucible.

Verifying the metrics for characterizing the melt system will be achieved through controlled and repeatable testing with consistent feedstock types and operating parameters. Any remaining material in the crucible will be tracked across multiple cycles to show that it can be recovered over time which supports the previous testing results that near-total material recovery is achievable with continuous operation.

The plastic pyrolysis system generates three primary output types: liquid oils, gaseous products, and solid byproducts. This characterization plan will quantitatively evaluate these outputs, verify their consistency, and demonstrate that the system properly converts plastic material into usable products. Oils produced from each cycle of pyrolysis will be collected downstream. The yield will be quantified through mass and volume measurements, which helps the direct calculation of conversion efficiency relative to the input plastic. In addition to the yield, other physical observations such as color, viscosity, and phase stability will be documented to assess product quality and repeatability. Additional experimental runs using the same input plastic and operating conditions will be performed to ensure that the oil production is consistent and reproducible. This would suggest that the observed outputs are intrinsic to the system rather than the result of isolated operating conditions. These were observed and recorded after successfully testing the plastic inputs shown in Figure 10.

Gaseous products will be characterized by total volume generated, flow rate, and general composition. The pyrolysis gases will be pushed through a water bubbler, which leads into calibrated collection vessels to allow for accurate volumetric measurements. The gas liberated from the recent Milestone system test is shown in Figure 11 venting into a fume hood. Gas sensors will be



Figure 10: 700 grams of polyethylene used for testing.

employed to detect the hydrocarbons and track any changes in composition across the test runs. Repeated measurements over multiple cycles will allow trends and composition in gas yields to be identified, which will help to evaluate process stability and the effectiveness of the pyrolysis system in producing valuable gaseous outputs.



Figure 11: Syngas generated from pyrolysis reaction.

Solid byproducts such as char or residual material remaining in the reaction chamber (as depicted in Figure 12) will be characterized by mass and qualitative appearance. Measuring the mass of the solid residue after each run provides insight into the conversion completeness and carbon distribution in the system.



Figure 12: Plastic char generated from pyrolysis reaction.

Verifying these characterized outputs will be achieved through repeatable testing under controlled conditions. Multiple runs with consistent inputs will ensure that the measured outputs are reproducible and directly attributable to the recycling process. Quantitative yield measurements, observable physical characteristics, and repeatability provides an adequate framework for evaluation of the plastic pyrolysis.

2.3 Milestone Round Schematics

The overall design process is overviewed in Figure 13, which shows both the plastic and metal feedstock being heated from the same microwave energy source. The heat creates a reaction that converts the solid feedstocks into raw products which can later be refined. Syngas and char can be transformed into methane and concrete filling while molten aluminum can be recast into various parts.

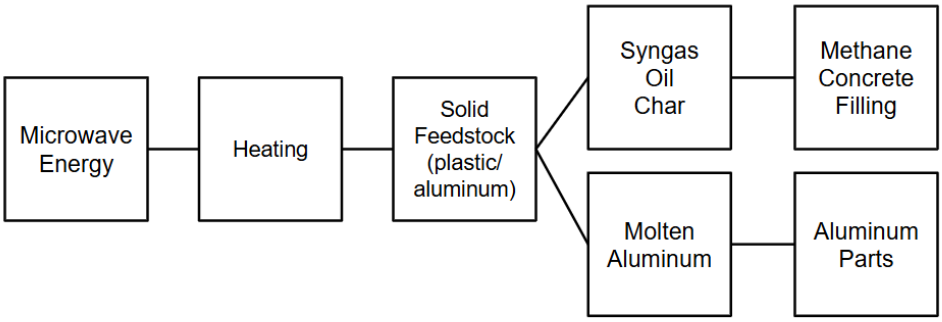


Figure 13: Design Process Schematic.

An flow of the microwave power of the Final Round RECLAIM system is shown in Figure 14. Power is first converted and fed to a solid-state microwave generator. The SSMG generates 2.45-GHz microwaves and transmits them to a microwave isolator. After leaving the isolator, the microwaves travel through a tuner and into the reaction chamber where they are absorbed by the feedstock. The various microwave components of the system are connected via WR340 waveguides.

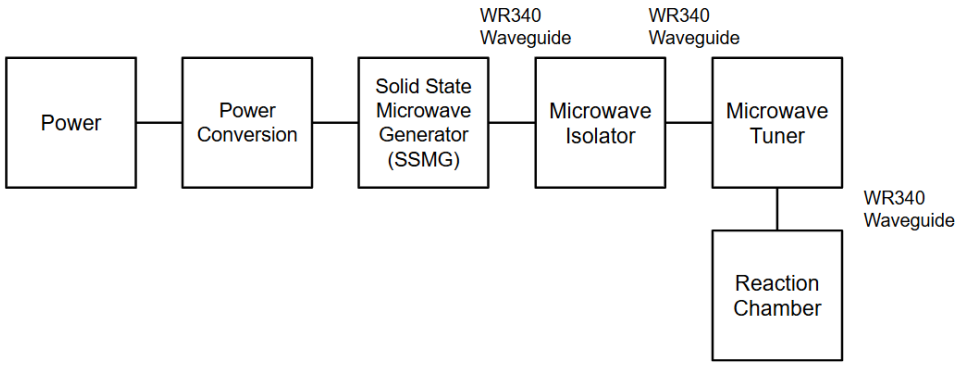


Figure 14: Electrical Schematic

Figure 15 shows the developed and tested Milestone microwave plastic pyrolysis system. The reaction begins in A (the reaction chamber). The produced gas then flows into B (the cooling jacket), where the condensable fraction is cooled and collected as oil at C. Non-condensable gases are released at D.

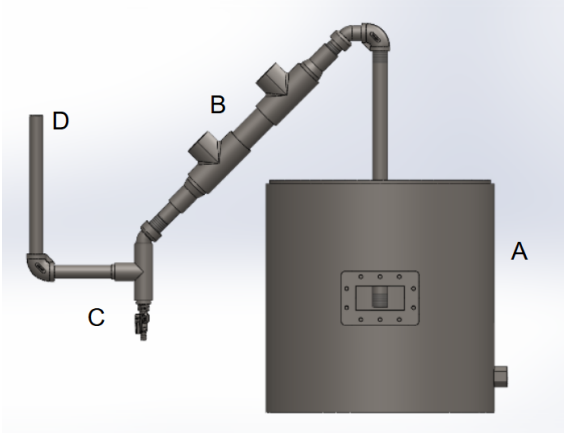


Figure 15: Front view of the proof-of-concept microwave plastic pyrolysis system.

Figure 16 shows the current proof-of-concept smelt system. Microwaves are first processed by A (the Daihen CMC-10A Automatic Microwave Tuning Control Unit). The signal then passes through a WR340 isolator and into B (the Daihen SMA-15B tuner). From there, the microwaves enter C (the reaction chamber), where the metal is melted.

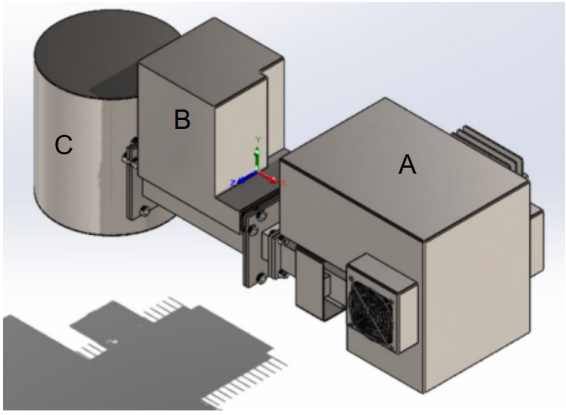


Figure 16: Isometric view of the current smelt proof-of-concept system.

The overall schematic of RECLAIM was introduced with showing the design and electric process. Figures of the plastic pyrolysis and smelt system have also been shown. These currently separate parts will be combined to one system.

2.4 Milestone Round Master Equipment List

Description of Equipment and Supplier	Estimated Mass (kg)	Estimated Volume (cm^3)
Crucible	2–3	336.5
Resonance Chamber (Houses Crucible)	10	21230
Reaction Chamber (Houses Plastic)	10	21230
Mass Flow Controller (MKS 1579A)	2.7	1046
Multi Gas Controller (MKS 647B)	8.2	15256
Magnetron Controller Power Supply (Daihen ATP-10A)	15.9	27788
Magnetron 2.45 GHz (Daihen ATM15a)	7.5	4720
Microwave Tuning Unit (Daihen CMC-10A)	2.3	6293
Daihen SMA15B Tuner	13.6	5529
Isolator (WR340)	4.9	4670
Waveguide Connector (WR340) (×2)	1.4 (Combined)	2596 (Combined)
90° Waveguide Elbow (WR430)	2.5	2655
WR 430 Waveguide	2	743
WR 374 Waveguide	2	712
DR-75-24 (Waveguide connector)	1	246
3/4″ Black Steel Pipe (18″ & 24″ lengths)	1.8	298
1-1/4″ Black Steel Pipe (12″ length)	1.0	426
1/2″ Black Steel Pipe (6″ & 10″ lengths)	0.3	73
Steel Elbows & Fittings (Various)	0.5	210
Argon Cylinder (Size T)	66	60567
Water Hose x 2	5.4	350
Gas Hose	4	150

2.5 Milestone Round Safety Analysis

2.5.1 Identify any potentially hazardous materials or other safety concerns related to your design, concept of operations, or outputs

The proposed system incorporates two subsystems: a plastic pyrolysis unit and a metal recycling and casting unit. Both subsystems involve high temperatures, reactive materials, and potentially hazardous byproducts. Identifying these hazards is essential to ensuring safe operation during testing and demonstration and to informing appropriate safety procedures. The plastic pyrolysis subsystem presents several hazards. During pyrolysis, plastics are heated to high temperatures in the absence of oxygen, producing a mixture of gases and condensable hydrocarbon products. The pyrolysis gases include flammable hydrocarbons such as methane, ethane, and other light hydrocarbons, as well as carbon monoxide and carbon dioxide. Many of these gases pose fire or health risks if released into enclosed spaces. Additionally, hot pyrolysis vapors and condensed oils can cause burns upon contact, and some liquid products may contain aromatic or oxygenated compounds that are hazardous if inhaled or handled without appropriate protection. Microwave

heating introduces additional safety concerns specific to electromagnetic energy and thermal exposure. Improper interaction between microwave energy and the reactor contents could lead to hot spots or arcing within the chamber. These effects can damage system components and pose a burn or fire hazard. The reactor and surrounding structures may reach elevated temperatures during operation, increasing the risk of thermal injury if accessed while hot. Electrical hazards associated with the microwave power supply and wiring are also present if components are improperly protected or handled. The aluminum recycling subsystem introduces hazards associated with molten metal processing. Aluminum must be heated above its melting point, resulting in a liquid metal at temperatures exceeding 660 °C. Molten aluminum presents a significant burn hazard, and accidental contact can result in injury. An uncontrolled pour of molten aluminum or blockage in the melt flow path could also lead to splashing or pressure-related hazards. The use of inert argon gas in the aluminum subsystem presents further safety considerations.

While argon is non-toxic and non-flammable, it is an asphyxiation hazard in confined or poorly ventilated spaces due to oxygen displacement. Overall, the primary hazardous materials and safety concerns associated with the design include flammable pyrolysis gases, hot and potentially toxic condensable liquids, microwave energy exposure, molten aluminum, inert gas displacement, and high-temperature system components. These hazards are inherent to thermal recycling and metallurgical processes and must be managed through appropriate design choices, operating procedures, and safety controls during system demonstration and use.

2.5.2 Safety analysis addressing pressure vessel systems and material compatibility (if relevant to your solution)

Material compatibility was deeply rooted in the decision-making process for both the pyrolysis section and smelting system. Smelting utilized a stainless steel reaction chamber due to its high thermal stability. Silicon carbide (SiC) powder was used to create a crucible that would absorb the microwave energy and heat up the metal within. The SiC crucible was wrapped with ceramic fiber blankets and Kapton tape to insulate the chamber. This material selection was essential for the safety and efficiency of the solution.

The pyrolysis system utilizes a similar chamber. The material used for this subsystem is made of steel, stainless steel, and iron. These materials were chosen due to withstand the high temperatures from the gas and oil produced from the reaction. The cooling jacket for the pyrolysis section used black steel because of its higher thermal conductivity rather than stainless steel.

The inflow of Argon creates the inert atmosphere within the chamber, but also subsequently pressurizes the chamber when flow is constrained by the collection device. The current system only collects small samples from the outlet at measured periods of time; when it is not the flow is safely vented into the building's ventilation system. When dealing with pressurized systems, it's necessary to take caution and ensure proper equipment is utilized. The Mass Flow Controller, alongside the Multi-variable Gas Controller are used to ensure that the flow of argon into the chamber is consistent and safe. The system was verified to be airtight before testing for a reaction. Maintaining gas containment and controlled venting is therefore critical during operation. Across both subsystems, high-temperature components, sealed chambers, and interconnected piping introduce general safety concerns related to system integrity and reliability. Gaskets, seals, and joints exposed to repeated thermal cycling may degrade over time, increasing the risk of leaks. During testing, leaks became present due to thermal stress; testing was halted shortly after this. While small leaks don't present a significant risk, especially inside a fume hood, system integrity began to quickly deteriorate. Ensuring volatile gases and oxygen did not mix within the reaction chamber was ensured. It is a high risk that was directly addressed through the system design and verification testing.

3 Development/Project Plan

RECLAIM in the milestone round demonstrates the capability of melting metals and pyrolyzing plastics. Further development of RECLAIM for the final round system will center around reducing size and increasing the performance. The final design will integrate both systems into a single reaction chamber. To account for future moon missions, RECLAIM is being sized for standards similar to the ISS "Express Rack" system [8]. While standard payload accommodations for future lunar missions have not yet been finalized, designing RECLAIM to fit within current standards demonstrates its compatibility with a standard payload structure once such specifications are established. The Express Rack dimensions are 2 m (79 in) tall, 1.05 m (41 in) wide, and 85.9 cm (33.8 in) deep with an undefined mass limit. We presented our milestone round system within the Express Rack outer support structure, showing significant volume remaining for all supporting systems (Figure 17). Our system is mass-conservative and expected to meet the requirements set forth for the final round. The integrated dual-process architecture is particularly advantageous due to its ability to recycle multiple waste streams. The final design iteration prioritizes low mass, reduced volume, and improved energy efficiency—key attributes for a viable lunar recycling system.

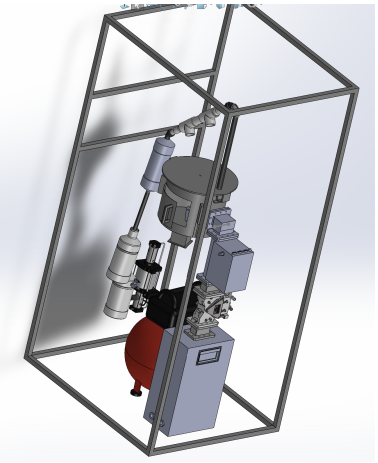


Figure 17: CAD model of RECLAIM system within the envelope of an Express Rack.

To help lower weight and improve efficiency , the old power system was redesigned. The current microwave power system is larger than necessary and can be reduced in size to improve transportability. For the final round, the RECLAIM system will replace the magnetron source with a Solid State Microwave Generator (SSMG) which connects to a WR340 isolator and HOMER autotuner basic unit model STHT 1.8 tuner. This new system alone cuts down on 8,300 cubic centimeters.

In addition to the power system, the reaction chamber will be modified to allow for two separate processes to occur in the same chamber with minimal astronaut interaction. A door on the side of the chamber was introduced to allow for easy access into the chamber so the crucible and plastics can be swapped in and out effortlessly. We've designed the door to be airtight to minimize leakage of any microwaves or gas. Replacing the top lid and the slow tedious and unreliable method of clamping and lowering the work and increasing the safety for the user.

A new casting system is integrated within the system as a whole. The crucible sits above a drain hole that remains plugged with an alumina pipe. After completion of a smelting cycle, the pipe can be lifted out of the hole, allowing the molten aluminum to drain out of the reaction chamber, channeled through another pipe into a casting or mold. This not only helps speed up the casting process, but also increases the safety for the user by avoiding a manual pour. The reaction chamber also includes an inlet for an inert gas and an outlet for access to the other pyrolysis subsystems. These will be sealed off during smelting using a valve system. The casting system will also be removed during the pyrolysis reaction, and actuator-based seals will be used to ensure air-tightness.

While the current pyrolysis subsystems function as intended an entire redesign of the system was required to meet volume constraints and ensure system integrity. Originally, NPT fittings were used for the gas, pipes and for the cooling jacket. While offering a cost-effective system design, Swagelok fittings are safer and more reliable for the pyrolysis subsystem. The introduction of a catalyst jacket, which may be regeneratively or externally heated, an oil filter, a larger oil collection chamber, a water bubbler, a gas booster, a condenser, and a gas tank will also be added.

The current proof-of-concept milestone system cannot collect all of the byproducts, only samples, generated from the reaction. The final round design addresses this and increases the generation of the desirable byproducts, i.e., methane and other volatiles.

A catalyst jacket could be introduced directly above the chamber for improving the pyrolysis reaction. It has been shown that routing the gaseous products created during the pyrolysis reaction through a heated catalyst at around 800°C can increase the yield of non-condensable hydrocarbons [9]. A promising catalyst that could be added to the jacket process is ilmenite ($FeTiO_3$). Ilmenite is found on the Moon, so would be readily accessible on the lunar surface. It aids in the production of H_2, CO, and other volatile hydrocarbon gases.

The pyrolysis cooling jacket will be redesigned to ensure more efficient cooling. After the cooling jacket, a filter for both gases and oil will be introduced to remove any undesirable toxic chemicals. The oil collection unit will also be introduced to capture any oil generated through the cooling jacket. Finally, syngas will be routed through a water bubbler, which further cools the volatile gas, ensuring proper system flow and safe storage. It will than be collected and a gas booster and condenser will route the gas into a collection tank where it can be stored and later utilized for practical applications such as the creation of different propellants, fuels, or heating sources.

Figure 18 below depicts a timeline for the final round submission. We recognize that development will need to occur quickly and over the summer.

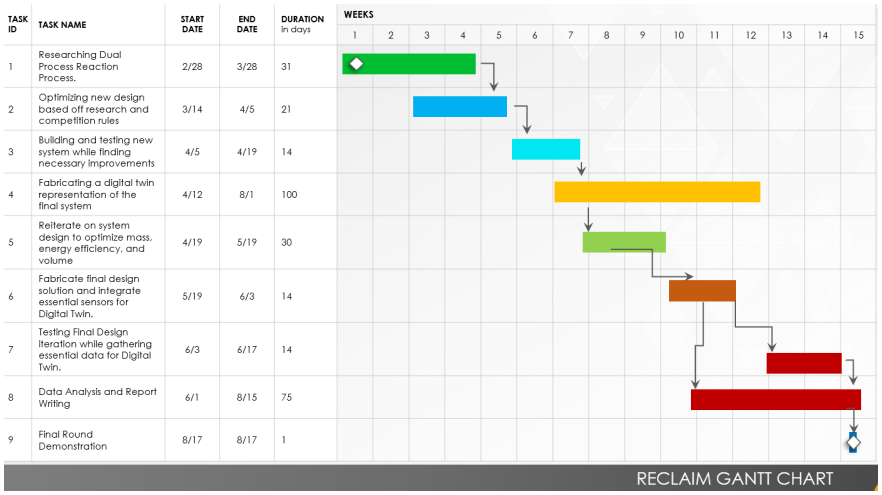


Figure 18: Timeline for RECLAIM Final Round development.

4 Risk Assessment

A. High Temperature Byproducts

High temperatures generated from the microwave source present a medium risk during operation. Pyrolysis and smelting require high temperatures to complete their respective recycling process. Operating around such high temperatures presents risk for burns or damage to surrounding equipment. For personnel, this risk is controlled by limiting human interaction during active operation, following proper procedures, and requiring appropriate protective equipment when handling system components. To minimize exposure, system access points are clearly defined, safety interlocks added, and operating procedures developed during testing are followed. Additionally, temperature is constantly monitored closely to ensure safety and efficiency; high heat sources are

insulted; and conduction paths minimized.

B. Flammable Gases

The production of flammable gases during pyrolysis introduces a medium to high risk, namely in containment and pressure management. Plastic pyrolysis generates volatile gases that must be safely transported, cooled, and collected. Leaks and pressure fluctuations could cause safety concerns if not properly controlled. To eliminate this risk, the system is designed with sealed steel piping (e.g., Swagelok) and a cooling jacket to reduce byproduct temperature. The chamber has an inert atmosphere produced by the flow of argon and controlled with a multi-gas controller that ensures the system pressure within the chamber is always greater than the atmospheric pressure, ensuring no oxygen will enter.

C. Sealing and Handling the Gas

Keeping the system airtight and handling the gas reliably is also a medium risk. Maintaining an airtight environment is essential for stable and efficient pyrolysis and preventing unwanted combustion. During testing, epoxy sealing methods were established. As the system continues to be developed, regular inspection of seals and connections will be conducted, and components subject to high stresses, such as thermal effects, will be replaced as needed. These measures help maintain system integrity during demonstrations.

D. Catalyst Performance and Material Buildup

Catalyst performance and material buildup in the cracking and cooling stages is also considered to be a medium risk. Over time, catalysts may degrade, which could reduce hydrocarbon breakdown efficiency or restrict gas flow. This risk is minimized by designing the catalyst section to be easily accessible for inspection and replacement. Proper operating conditions identified during testing will be maintained to ensure components remain operational for as long as intended.

E. Logistics and Component Reliability

It is important to consider logistical and component reliability risks when preparing the system for the Final Round. This is a low to medium risk with massive implications, as components such as heating elements, sensors, and catalysts may require replacement or recalibration before demonstration. To address this risk, commonly available materials are used wherever possible, and spare components are planned to ensure continued operation if unexpected problems occur. Risk assessment is critical in the development and demonstration of the microwave-assisted plastic and aluminum recycling system.

F. Microwave Issues

The operation of the pyrolysis reactor relies on effective coupling between microwave energy and the contents of the microwave chamber. Variations in feedstock composition and geometry may result in uneven microwave absorption, leading to non-uniform heating or localized hot spots. In addition, improper interaction between microwaves and chamber components could result in microwave reflection or arcing. This risk is assessed as medium, as it could affect both performance and system safety. To mitigate this risk, a cylindrical reactor geometry was selected to provide more uniform deposited energy distribution.

G. Incomplete Condensation

After exiting the reaction chamber, pyrolysis vapors remain at elevated temperatures and may exhibit high flow rates. Under these conditions, incomplete condensation could occur, reducing liquid product recovery and allowing condensable hydrocarbons to pass into downstream gas-handling components. This risk is assessed as low to medium. To address this risk, the system is equipped with an external cooling jacket, which promotes rapid cooling and effective condensation of pyrolysis vapors into liquid products. Table 3 goes through the risks involved for smelting and plastic pyrolysis.

Table 3: RECLAIM Risk Assessment Table.

Risk	Cause	Consequence	Mitigation
Thermal Hazards	Overheating of crucible or molten material handeling	Equiptment damage, burn hazards, system downtime	Use backup crucibles and controlled heating
Flammable Gases	Generation and leakage of hydrocarbon vapors during pyrolisis	Fire or explosion risk, personnel injury	Use seals, pressure monitoring, and leak checks
Gas Sealing and Handling	Inadequate sealing or component wear or improper connections	Loss of system efficiency, combustion risk, exposure hazards	Inspect seals regularly and apply proper sealing methods
Condensation Inefficiency	High vapor temperature or flow rate reducing condenser effectiveness	Reduced liquid recovery, downstream fouling	Use a water-cooled condenser with sufficient time and controlled flow rates
Material Compatability	Chemical interaction between molten aluminum and incompatible materials	Component degradation, blockage, reduced reliability	Select high-temperature compatible materials and inspect seals and contact surfaces
Component Reliability	Failure or drift of hearters, sensors, or catylysts	Performance degradation or test delays	Use standard components, maintain spares, and perform pre-test checks

5 Development Budget

As shown in Table 4, we estimate a total budget of $185k to complete the Development Plan in the Final Round. This budget assumes the $50k Milestone Round award; $25k in internal funds and fundraising from SSPL/Penn State alumni and corporations; and $110k of leveraged internal resources available in Bilén's lab. Our experience with similar past efforts (e.g., NASA's BIG Idea Challenge, NASA's 3D-Printed Habitat Challenge) provides experience for this costing effort. If required components are not already available from Bilén's lab, much of the required components can be purchased from eBay, as they have been for past challenges.

Table 4: RECLAIM Budget

Type of Cost	Description	Estimated budget for Final Round	Expected funding source(s)
Materials	Hardware, structural metal, electrical components, fabrication costs	$25k	Internal funds and fundraising
	Controls, sensors, wire, hardware, tubing, fittings, etc.	[$25k]	Available from lab
	Microwave Components	[$25k]	Loaned and available from lab
Equipment	SSMG	[$60k]	Loaned (available from lab)
Lab/Testing	Gases and expendable materials	$5k	Milestone Round Award
Personnel	4 summer interns	$5k stipends, $20k total	Milestone Round Award
Travel	6 people	$2k per traveler, $12k total	Milestone Round Award
Admin	15%	$6k	Milestone Round Award
Other (as applicable)	Reserve	$7k	Milestone Round Award
	Totals	$75k and [$110k]	

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